12.16 Markov Chains Forwards and Backwards

WORKSHEET

The question

STEP III 1996, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/96-s3-q12

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself What does the condition "each row adds up to 1" mean? Do the columns add up to 1?

STAGE 1 One step

Covers: Part (i)

Ask yourself Condition on the type of the current theorem and use the law of total probability.

STAGE 2 The stationary distribution

Covers: Part (ii)

Pause and try Write out $a_j = a_1 p_{1j} + a_2 p_{2j} + a_3 p_{3j}$ for $j = 1, 2, 3$, with $a_1 + a_2 + a_3 = 1$. Which equation is simplest to start with?

STAGE 3 Running the chain backwards

Covers: Part (iii)

Ask yourself q_{ij} is the probability that the preceding theorem was of type i, given that this one is of type j. Use Bayes' theorem, with your answer to Part (ii) for both theorems.

What's really going on

Convergence to equilibrium

Start Professor Ecks with any theorem; after n more, the distribution of the type is $\mathbf{a}_0 P^n$. For a chain like this one, in which every type can follow every other, P^n converges to a matrix with every row equal to the stationary distribution $(\frac{1}{6}, \frac{1}{3}, \frac{1}{2})$, whatever the starting point. The speed of convergence is governed by the other eigenvalues of P (here 0.2 and 0.1), which shrink geometrically: after just a few theorems, her past is forgotten. This *ergodic theorem* for Markov chains is a central result of a second-year probability course, and it is what makes Markov chain Monte Carlo methods work: to sample from a complicated distribution, design a Markov chain with that stationary distribution, and run it for a long time.

Reversibility

The backwards chain (q_{ij}) is different from the forwards chain (p_{ij}): you can tell the direction of time from a long record of her theorems. Chains whose backwards and forwards versions agree, $a_i p_{ij} = a_j p_{ji}$ for all i, j, are called *reversible*; they include random walks on networks and the models of physical systems in thermal equilibrium. Reversibility is a powerful property, because it makes the stationary distribution easy to find.

Markov and Pushkin

Andrey Markov introduced his chains in 1906, partly to show that the law of large numbers does not require independence, in a dispute with the mathematician Pavel Nekrasov, who had claimed it did. In 1913 Markov applied his idea to the first 20 000 letters of Pushkin's verse novel *Eugene Onegin*, classifying each as a vowel or consonant

and showing that the type of each letter depends on the type of the previous one in a way well described by a two-state chain. It was the first statistical analysis of a text, and the ancestor of every predictive-text system.

Try this next

1 **A rabbit warren.** Try STEP III 2003, Question 13 (step.maths.org/questions/03-s3-q13): a random walk on the corners of a square with an exit at each corner, solved by first-step equations.

2 **Two steps.** If Professor Ecks's last theorem was new and correct, find the probabilities of the three types for the theorem after next.

3 **Eigenvalues.** Show that the eigenvalues of P are 1, 0.2 and 0.1.

4 **Reversible?** Check whether $a_i p_{ij} = a_j p_{ji}$ for Professor Ecks's chain. What does the answer tell you?

Further reading

- **James Norris, *Markov Chains*** (Cambridge University Press). The standard undergraduate text.
- **Olle Häggström, *Finite Markov Chains and Algorithmic Applications*** (Cambridge University Press). Short, and good on Markov chain Monte Carlo.
- **Brian Hayes, "First links in the Markov chain"**, *American Scientist* 101 (2013). Markov's Pushkin analysis, retold.

The full guided solution is in Section 12.16:
https://beyondstep.keithmcnulty.org/ch12/markov.html